

A UNIFORM ORTHOGONAL FILTRATION ON HIGHER CHOW GROUPS

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ABSTRACT. Let X/k be a smooth projective variety of dimension d . We construct a canonical filtration $F^\bullet \mathrm{CH}^n(X, m)_R$ on the higher Chow groups of X that is uniform in the Adams degree m . The key input is that the object

$$B_{n,m} := M(X)(-n)[-(2n-m)]$$

lies in an effective orthogonal window $0 \leq r \leq d-n$ independent of m . This yields a strongly convergent spectral sequence

$$E_{r,q}^1(n, m) \Rightarrow \mathrm{CH}^n(X, m)_R$$

whose column range does not depend on m , extending the orthogonal construction of Galindo–Pelaez from ordinary Chow groups to all higher Chow groups. Strong convergence is verified explicitly by checking Boardman’s condition $\mathrm{RE}\infty$ using the finiteness of the column range.

We then identify geometric situations in which the filtration is computable and satisfies the Bloch–Beilinson axioms. For products of $d \geq 2n$ curves and for the smooth cubic threefold, the spectral sequence degenerates and the resulting filtration satisfies all five axioms unconditionally. For abelian varieties, degeneration is forced by Beauville eigenspace separation or by Deninger–Murre projectors; the algebraicity of these projectors over the base field k is established explicitly via Lagrange interpolation of multiplication-by- N correspondences. This gives all five axioms unconditionally for $n = 1$, for all n over finite fields, for all $n \leq g/2$ over $\mathbb{C}$ or whenever $\rho(A) = 1$, and—new to this version—for CM abelian varieties over imaginary quadratic fields and over function fields of curves over $\mathbb{F}_q$.

We extend the realization theory to include full compatibility with mixed Hodge structures over $\mathbb{C}$: the orthogonal filtration is shown to refine the Hodge coniveau filtration, each graded piece lands in the correct Hodge-type component, and the Abel–Jacobi map factors compatibly through the Beauville decomposition of the intermediate Jacobian.

Over number fields the decisive phenomenon is instead non-splitting: the associated motivic tower need not split even when the spectral sequence degenerates. We give a complete proof that the obstruction is an Ext^1 -class detecting Mordell–Weil and Griffiths-type phenomena, thereby isolating the precise point at which the Bloch–Beilinson philosophy becomes arithmetic.

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2020 *Mathematics Subject Classification*. 14C25, 14F42, 19E15.

Key words and phrases. higher Chow groups, motivic cohomology, orthogonal truncations, weight structures, abelian varieties, Bloch–Beilinson filtration.

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1. INTRODUCTION

The central question. A codimension- n algebraic cycle on a smooth projective variety X defines a class in $\mathrm{CH}^n(X, m)$. The Bloch–Beilinson philosophy predicts that these groups carry a canonical filtration whose successive steps reflect increasingly subtle realizations: cycle classes, intermediate Jacobians or regulators, and ultimately special values of L -functions [8, 23, 32]. The problem is to construct such a filtration directly in the motivic category, to prove that it is uniform in the Adams degree, and to understand when it is geometric and when it is genuinely arithmetic.

The orthogonal tower of Pelaez and Galindo–Pelaez [34, 21] already gives a candidate when $m = 0$. The main structural advance of the present paper is that the same method extends to all higher Chow groups without changing the effective window. More precisely, for

$$B_{n,m} = M(X)(-n)[-(2n - m)],$$

the cohomological shift does not alter the orthogonal range: the Tate levels remain $0 \leq r \leq d - n$, independently of m (Proposition 2.2). From this we obtain a strongly convergent spectral sequence and a filtration $F^\bullet \mathrm{CH}^n(X, m)_R$ whose length is bounded by $d - n + 1$ uniformly in m (Theorem A).

What is new. Relative to [21], the main new ingredient is Lemma 2.7, which shows that for the input object $B_{n,m} = M(X)(-n)[-(2n-m)]$ the cohomological shift is invisible to the orthogonal tower. This is what makes the passage from $m = 0$ to all $m \geq 0$ genuinely uniform rather than a reformulation.

The structural advance unfolds in three layers corresponding to the trichotomy (existence / degeneration / non-splitting): Theorem A gives canonical existence; Theorems C, E, and 9.9 give geometric degeneration in explicit families; Theorem 11.4 isolates the arithmetic obstruction to splitting, identifying it with Ext^1 -classes detecting Mordell–Weil and Griffiths-type phenomena. Theorem B records the functoriality, and Theorem D assembles the BB-axiom verification for abelian varieties including the unconditional range of Theorem 10.2.

The main theorems.

Theorem 1.1 (A). *Let X be smooth projective of dimension d , and $n \geq 1$, $m \geq 0$. There is a strongly convergent spectral sequence $E_{r,q}^1(n, m) \Rightarrow \text{CH}^n(X, m)_R$ with*

$$E_{r,q}^1(n, m) \cong H_M^{2n-m+q-r, n+r}(X, R)$$

and column range $0 \leq r \leq d - n$ independent of m .

Theorem 1.2 (B). *If $\alpha \in F^r \text{CH}^n(X, m)_R$ and $\beta \in F^s \text{CH}^{n'}(Y, m')_R$, then $\alpha \times \beta \in F^{r+s} \text{CH}^{n+n'}(X \times Y, m + m')_R$. If $i: Z \hookrightarrow X$ has codimension c and $\alpha \in F^r \text{CH}^{n-c}(Z, m)_R$, then $i_*(\alpha) \in F^{r+c} \text{CH}^n(X, m)_R$.*

Theorem 1.3 (C — Degeneration and Beauville splitting). *For A/k an abelian variety of dimension g over any perfect field, the orthogonal spectral sequence degenerates at E^1 and*

$$\text{gr}_F^r \text{CH}^n(A, m)_{\mathbb{Q}} \cong \text{CH}_{(n+r)}^n(A, m)_{\mathbb{Q}} \quad \text{for } 0 \leq r \leq g - n.$$

Theorem 1.4 (D — BB filtration for abelian varieties). *For A/k abelian of dimension g , the filtration $F_{\text{orth}}^j = F^{n-j} \text{CH}^n(A)_{\mathbb{Q}}$ satisfies all Jannsen axioms [23]. Axiom (BB2) holds: unconditionally for $n = 1$; for all n over $\mathbb{F}_q$; for all $n \leq g/2$ over $\mathbb{C}$ or when $\rho(A) = 1$ (Theorem 10.2). The remaining open case is $n \geq 2$, $\rho(A) \geq 2$, over a non-finite non-complex field.*

Theorem 1.5 (E — Unconditional BB for products of $d \geq 2n$ curves). *For $X = C_1 \times \cdots \times C_d$ with $d \geq 2n$ smooth projective curves over any perfect field, all five axioms (BB1)–(BB5) hold unconditionally for all $n \geq 1$. The top Beauville piece has rank $\binom{d}{n}$.*

Organisation. Sections 2–4 prove Theorems A and B. Section 5 tests the filtration with six examples. Section 6 proves Theorem C. Sections 7–8 prove Theorem D and rank formulas. Section 9 proves Theorem E and Theorem 9.9. Section 10 proves Theorem F and the Frobenius degeneration example. Section 11 collects comparisons: AHSS, Bloch–Ogus coniveau, Ext-length and Theorem 11.4, and 1-motives. Section 12 treats realizations. Section 13 isolates limitations and directions.

2. PRELIMINARIES AND THE FILTRATION

2.1. Definitions and notation. We fix a perfect base field k of exponential characteristic p and set $R := \mathbb{Z}[1/p]$. We write Sch_k for k -schemes of finite type, Sm_k for smooth k -schemes

of finite type, and SmProj_k for smooth projective k -schemes. We work in Voevodsky's triangulated category

$$\mathcal{T} := \mathbf{DM}(k; R),$$

and we follow the notation of [21, §2]. Thus $\mathbf{1}_R := M(\mathrm{Spec} k)$ denotes the tensor unit, $[1]$ denotes suspension, (1) denotes the Tate twist, and $M(X)$ denotes the motive of X [28, Lec. 8, Lec. 14]; see also [16]. For $c \in \mathbb{Z}$ we write $A(c) := A \otimes \mathbf{1}_R(1)^{\otimes c}$.

The compact generators are

$$G_{\mathbf{DM}} := \{M(X)(p) : X \in \mathrm{Sm}_k, p \in \mathbb{Z}\}, \quad G^{\mathrm{eff}}(n) := \{M(X)(p) : X \in \mathrm{Sm}_k, p \geq n\},$$

and we write

$$\mathbf{DM}^\perp(n) := \mathrm{Loc}(G^{\mathrm{eff}}(n))^\perp.$$

This is the indexing used for the orthogonal tower in [21, §2.3–§2.5]. For smooth projective X one has the standard representability isomorphisms

$$(1) \quad \mathrm{Hom}_{\mathbf{DM}(k; R)}(M(X), \mathbf{1}_R(n)[2n - m]) \cong \mathrm{CH}^n(X, m)_R,$$

$$(2) \quad \mathrm{Hom}_{\mathbf{DM}(k; R)}(M(X)(-n)[-(2n - m)], \mathbf{1}_R) \cong \mathrm{CH}^n(X, m)_R.$$

2.2. Bloch–Beilinson filtration axioms. The Bloch–Beilinson–Murre conjecture [8, 23, 32] predicts a canonical decreasing filtration $F^\bullet \mathrm{CH}^n(X)_\mathbb{Q}$ satisfying five axioms (BB1)–(BB5) stated in Convention 2.6 and Section 8.

2.3. The orthogonal tower in $\mathbf{DM}(k; R)$. We consider the decreasing family of compact generators $\{G^{\mathrm{eff}}(n)\}_{n \in \mathbb{Z}}$ in the sense of [21, §2.3].

Definition 2.1. For $s \in \mathbb{Z}$ let

$$j_{s+1} : \mathbf{DM}^\perp(s+1) \hookrightarrow \mathbf{DM}(k; R)$$

be the inclusion and let

$$p_{s+1} : \mathbf{DM}(k; R) \rightarrow \mathbf{DM}^\perp(s+1)$$

be its right adjoint, whose existence follows from Neeman's theorem [26, Thm. 4.1]. Following [21, §2.3.1], the orthogonal cover is

$$\mathbf{bc}_{\leq s} := j_{s+1} \circ p_{s+1}.$$

The graded piece $\mathbf{bc}_{s/s-1}$ is defined by the canonical distinguished triangle

$$\mathbf{bc}_{\leq s-1} A \longrightarrow \mathbf{bc}_{\leq s} A \longrightarrow \mathbf{bc}_{s/s-1} A \longrightarrow \mathbf{bc}_{\leq s-1} A[1].$$

Proposition 2.2 (Orthogonal window for higher Chow groups). *Let $X \in \mathrm{SmProj}_k$ of dimension d , $n \geq 1$, $m \geq 0$, and set $B_{n,m} := M(X)(-n)[-(2n - m)] \in \mathbf{DM}(k; R)$.*

- (a) $B_{n,m} \in \mathbf{DM}^\perp(d - n + 1)$.
- (b) $\mathbf{bc}_{\leq d-n}(B_{n,m}) \xrightarrow{\sim} B_{n,m}$.
- (c) The orthogonal spectral sequence attached to $(B_{n,m}, \mathbf{1}_R)$ has effective column range $0 \leq r \leq d - n$. This range is independent of m .

Proof. We apply the abstract window result [21, Prop. 3.1, Cor. 3.2] with $Y := X$, $e := d$, $a := -n$, $t := -(2n - m)$, giving $N = B_{n,m} = M(X)(-n)[-(2n - m)]$ and effective Tate level $e + a = d - n$.

(a) We must show $\mathrm{Hom}_{\mathbf{DM}}(M(Z)(d-n+1)[s], B_{n,m}) = 0$ for all $Z \in \mathrm{SmProj}_k$ and $s \in \mathbb{Z}$. By the cancellation theorem [28, Cor. 16.26]:

$$\mathrm{Hom}(M(Z)(d-n+1)[s], M(X)(-n)[-(2n-m)]) \cong H_{\mathbf{M}}^{-(2n-m)-s, -(d+1)}(Z \times X, R).$$

Motivic cohomology $H_{\mathbf{M}}^{p,q}(W, R)$ vanishes for $q < 0$ [28, Prop. 14.16]. Here $q = -(d+1) < 0$, so the group vanishes identically.

(b) Consider the exact triangle $\mathbf{bc}_{\leq d-n}(B_{n,m}) \rightarrow B_{n,m} \rightarrow P \rightarrow (\mathbf{bc}_{\leq d-n}(B_{n,m}))[1]$. By construction $P \in \mathrm{Loc}(G^{\mathrm{eff}}(d-n))$. From (a), $\mathrm{Hom}(K, B_{n,m}) = 0$ for any $K \in \mathrm{Loc}(G^{\mathrm{eff}}(d-n))$. The long exact sequence gives $P = 0$.

(c) The isomorphism in (b) implies $\mathbf{bc}_{\leq s}(B_{n,m}) \xrightarrow{\sim} B_{n,m}$ for all $s \geq d-n$, so the tower stabilises. Independence of m : the Tate level of $B_{n,m}$ is $a = -n$, giving $e + a = d-n$. The cohomological shift $t = -(2n-m)$ does not enter the Tate level. $\square$

Remark 2.3. The bigrading of $\mathbf{DM}(k; R)$ —Tate twist $(*)$ and cohomological shift $[*]$ —is essential. The orthogonal tower is indexed by Tate level; in a singly-graded triangulated category, changing m would change the window.

2.4. Tensor product stability of the orthogonal tower.

Lemma 2.4 (Tensor product stability). *In $\mathbf{DM}(k; R)$, for any $s, t \in \mathbb{Z}$:*

$$\mathrm{Loc}(G^{\mathrm{eff}}(s)) \otimes \mathrm{Loc}(G^{\mathrm{eff}}(t)) \subseteq \mathrm{Loc}(G^{\mathrm{eff}}(s+t)).$$

Consequently there is a canonical morphism

$$\mathbf{bc}_{\leq s} \mathbf{1}_R \otimes \mathbf{bc}_{\leq t} \mathbf{1}_R \longrightarrow \mathbf{bc}_{\leq s+t} \mathbf{1}_R.$$

Proof. If $M(Y)(a) \in G^{\mathrm{eff}}(s)$ and $M(Z)(b) \in G^{\mathrm{eff}}(t)$, then $a \geq s$ and $b \geq t$, hence

$$M(Y)(a) \otimes M(Z)(b) \cong M(Y \times Z)(a+b) \in G^{\mathrm{eff}}(s+t).$$

Since the tensor product is exact in each variable and preserves arbitrary coproducts, the inclusion extends from generators to the corresponding localizing subcategories. $\square$

2.5. The filtration.

Definition 2.5. *Let X be smooth projective of dimension d , and let $n \geq 1$, $m \geq 0$. For $0 \leq r \leq d-n$ define $F^r \mathrm{CH}^n(X, m)_R$ as the image of*

$$\mathrm{Hom}_{\mathbf{DM}}(B_{n,m}, \mathbf{bc}_{\leq -n+r} \mathbf{1}_R) \rightarrow \mathrm{Hom}_{\mathbf{DM}}(B_{n,m}, \mathbf{1}_R) \cong \mathrm{CH}^n(X, m)_R.$$

Set $F^{-1} := 0$.

Convention 2.6. The orthogonal filtration $F^\bullet$ is increasing. The Bloch–Beilinson filtration $F^\bullet$ is decreasing. The conversion throughout this paper is

$$F_{\mathrm{orth}}^j := F^{n-j} \mathrm{CH}^n(X, m)_{\mathbb{Q}},$$

so (BB1) reads $F^0 = F^n = \mathrm{CH}^n$ and $F^{n+1} = F^{-1} = 0$. Axiom (BB2) asserts $F^1 = F^{n-1} = \ker(\mathrm{cl}_\ell^n)$.

Lemma 2.7 (Twist compatibility). *For every $M \in \mathbf{DM}(k; R)$ and every integer c there is a natural isomorphism*

$$\mathbf{bc}_{\leq s}(M(c)) \cong (\mathbf{bc}_{\leq s-c} M)(c).$$

Proof. Step 1: generators. The autoequivalence $(c) = (-\otimes \mathbf{1}_R(c))$ maps generators: $M(Y)(a) \otimes \mathbf{1}_R(c) \cong M(Y)(a+c)$, so $(c)(\mathcal{G}_a) = \mathcal{G}_{a+c}$ and $(c)(\text{Loc}(G^{\text{eff}}(s))) = \text{Loc}(G^{\text{eff}}(s+c))$.

Step 2: the candidate $L := (\mathbf{bc}_{\leq s-c}M)(c)$.

- (i) $L \in \mathbf{DM}^\perp(s)$: apply (c) to $\mathbf{bc}_{\leq s-c}M \in \mathbf{DM}^\perp(s-c)$ and use Step 1.
- (ii) The cone of $L \rightarrow M(c)$ lies in $\text{Loc}(G^{\text{eff}}(s))$: the exact triangle (top row) twisted by (c) gives the cone in $\text{Loc}(G^{\text{eff}}(s))$ by Step 1.

By uniqueness of the right adjoint (Neeman [26, Thm. 4.1]), $\mathbf{bc}_{\leq s}(M(c)) \cong L$ naturally in M . $\square$

Remark 2.8 (The m -independence). Lemma 2.7 says precisely that Tate twist permutes the layers of the orthogonal tower. Applied to $B_{n,m} = M(X)(-n)[-(2n-m)]$, the filtration step

$$F^r \text{CH}^n(X, m)_R = \text{Im}(\text{Hom}(B_{n,m}, \mathbf{bc}_{\leq -n+r} \mathbf{1}_R) \rightarrow \text{Hom}(B_{n,m}, \mathbf{1}_R))$$

depends on r and n but not on m , because $\mathbf{bc}_{\leq -n+r} \mathbf{1}_R$ does not involve m .

3. THE SPECTRAL SEQUENCE

Lemma 3.1 (The E^1 -page). *Let X be smooth projective of dimension d , and let $n \geq 1$, $m \geq 0$. For $0 \leq r \leq d-n$,*

$$E_{r,q}^1(n, m) \cong H_M^{2n-m+q-r, n+r}(X, R).$$

Proof. By definition, $E_{r,q}^1 = \text{Hom}_{\mathbf{DM}}(B_{n,m}, \mathbf{bc}_{-n+r/-n+r-1} \mathbf{1}_R[q-r])$. Set $P_r := \mathbf{bc}_{-n+r/-n+r-1} \mathbf{1}_R$. Applying Lemma 2.7 to $B_{n,m} = M(X)(-n)[-(2n-m)]$ and using the tensor-hom adjunction

$$\text{Hom}_{\mathbf{DM}}(A(-c), B) \cong \text{Hom}_{\mathbf{DM}}(A, B(c)),$$

one obtains

$$E_{r,q}^1 = \text{Hom}_{\mathbf{DM}}(M(X)[-(2n-m)], P_r(n)[q-r]) \cong H_M^{2n-m+q-r, n+r}(X, R)$$

by [28, Lec. 14, Prop. 14.16]. The shift $-(2n-m)$ appears only in the cohomological degree; the Tate weight $n+r$ is independent of m . $\square$

Theorem 3.2. *The orthogonal tower on $\mathbf{1}_R$, truncated to $0 \leq r \leq d-n$, gives a strongly convergent spectral sequence*

$$E_{r,q}^1(n, m) \Rightarrow \text{CH}^n(X, m)_R, \quad 0 \leq r \leq d-n,$$

with differentials $d^\ell: E_{r,q}^\ell \rightarrow E_{r-\ell, q+\ell-1}^\ell$. The induced filtration on the abutment is $F^\bullet$ of Definition 2.5.

Proof. Set

$$A := \mathbf{1}_R, \quad B := B_{n,m} = M(X)(-n)[-(2n-m)].$$

With the notation of [21, §2.4], the orthogonal tower of A yields the spectral sequence

$$E_{p,q}^1 = \text{Hom}_{\mathbf{DM}(k;R)}(B, \mathbf{bc}_{p/p-1} A[q-p]) \Longrightarrow \text{Hom}_{\mathbf{DM}(k;R)}(B, A).$$

We first explain strong convergence. The object B is compact in $\mathbf{DM}(k;R)$, so Neeman's lemma gives

$$\text{Hom}_{\mathbf{DM}(k;R)}(B, \text{hocolim}_{p \rightarrow \infty} \mathbf{bc}_{\leq p} A) \cong \text{colim}_{p \rightarrow \infty} \text{Hom}_{\mathbf{DM}(k;R)}(B, \mathbf{bc}_{\leq p} A)$$

[26, Lem. 2.8]. Hence the induced filtration on the target is exhaustive. Moreover $B = M(X)(-n)[-(2n-m)]$ belongs to $\text{Loc}(G^{\text{eff}}(-n))$, while by definition $\mathbf{bc}_{\leq p}A \in \mathbf{DM}^\perp(p+1)$. Therefore

$$\text{Hom}_{\mathbf{DM}(k;R)}(B, \mathbf{bc}_{\leq p}A) = 0 \quad \text{for every } p \leq -n-1.$$

Equivalently, $E_{p,q}^1 = 0$ for all $p \leq -n-1$. Thus the spectral sequence is bounded on the left, and Boardman's criterion applies exactly as in [21, Thm. 4.1]; hence it converges strongly to

$$\text{Hom}_{\mathbf{DM}(k;R)}(B, \text{hocolim}_{p \rightarrow \infty} \mathbf{bc}_{\leq p}A).$$

It remains to identify the homotopy colimit with $A = \mathbf{1}_R$. The canonical map

$$c: \text{hocolim}_{p \rightarrow \infty} \mathbf{bc}_{\leq p} \mathbf{1}_R \longrightarrow \mathbf{1}_R$$

is an isomorphism in $\mathbf{DM}(k;R)$ by [21, Thm. 4.3]. Therefore the abutment is

$$\text{Hom}_{\mathbf{DM}(k;R)}(B, \mathbf{1}_R) \cong \text{CH}^n(X, m)_R$$

via (2).

Finally, Proposition 2.2(b) gives $\theta_{d-n}^{B_{n,m}}: \mathbf{bc}_{\leq d-n}(B_{n,m}) \xrightarrow{\sim} B_{n,m}$, so no column with $p > d-n$ contributes. Reindexing by $p = -n+r$ gives the finite range $0 \leq r \leq d-n$, and the induced filtration on the abutment is exactly the filtration of Definition 2.5. $\square$

Remark 3.3 (m -shift). $E_{r,q}^1(n, m+1) \cong E_{r,q-1}^1(n, m)$: passing $m \rightarrow m+1$ shifts cohomological degrees by -1 within unchanged columns.

4. BASIC FUNCTORIALITY

Theorem 4.1 (Product compatibility). *If $\alpha \in F^r \text{CH}^n(X, m)_R$ and $\beta \in F^s \text{CH}^{n'}(Y, m')_R$, then $\alpha \times \beta \in F^{r+s} \text{CH}^{n+n'}(X \times Y, m+m')_R$.*

Proof. The motivic Künneth isomorphism [28, Thm. 15.1] gives $B_{n+n', m+m'}(X \times Y) \cong B_{n,m}(X) \otimes B_{n',m'}(Y)$. Lifts $\tilde{\alpha}: B_{n,m} \rightarrow \mathbf{bc}_{\leq -n+r} \mathbf{1}_R$ and $\tilde{\beta}: B_{n',m'} \rightarrow \mathbf{bc}_{\leq -n'+s} \mathbf{1}_R$ tensor via Lemma 2.4 to a lift of $\alpha \times \beta$ at level $r+s$. $\square$

Theorem 4.2 (Regular Gysin shift). *Let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension c . If $\alpha \in F^r \text{CH}^{n-c}(Z, m)_R$, then $i_*(\alpha) \in F^{r+c} \text{CH}^n(X, m)_R$.*

Proof. The Gysin morphism gives $\gamma: B_{n,m}(X) \rightarrow B_{n-c,m}(Z)$ in $\mathbf{DM}(k;R)$. A lift $\tilde{\alpha}: B_{n-c,m}(Z) \rightarrow \mathbf{bc}_{\leq -(n-c)+r} \mathbf{1}_R$ composes with γ to give a lift of $i_*(\alpha)$ to $\mathbf{bc}_{\leq -n+(r+c)} \mathbf{1}_R$. $\square$

Remark 4.3. Under $F_{\text{orth}}^j = F^{n-j}$, Theorem 4.2 gives $i_*(F^j \text{CH}^{n-c}) \subseteq F^j \text{CH}^n$, in agreement with Jannsen's axiom (BB3).

5. EXAMPLES

Theorem 5.1 (Mixed Tate splitting). *If $M(X)_{\mathbb{Q}} \cong \bigoplus_{a \in I} \mathbf{1}_{\mathbb{Q}}(-u_a)[-v_a]$ in $\mathbf{DM}(k; \mathbb{Q})$, the orthogonal spectral sequence for $\text{CH}^n(X, m)_{\mathbb{Q}}$ degenerates at E^1 and $F^\bullet$ is canonically split.*

Example 5.2 ($\mathbb{P}^1 \times E$: m -independence made visible). Let E/k be an elliptic curve and $X = \mathbb{P}^1 \times E$, $n = 1$. The window is $0 \leq r \leq 1$, uniform in m . By Theorem A:

$$E_{0,q}^1(1, m) \cong H_M^{1-m+q,1}(X, R), \quad E_{1,q}^1(1, m) \cong H_M^{2-m+q,2}(X, R).$$

The two-column structure is unchanged as m varies; only the bidegree labels shift by m .

Example 5.3 (Non-split extension). Let E_1, E_2 be non-isogenous elliptic curves and $X = E_1 \times E_2$, $n = 1$. The extension class $\omega \in \text{Ext}_{\mathbf{DM}}^1(\text{gr}_F^1 B_{1,0}, \text{gr}_F^0 B_{1,0})$ records the integral non-splitting and is visible in $F^0 \subsetneq F^1$ even when the rational filtration splits.

Example 5.4 (Abelian fourfold, $n = 2$). Let A/k be an abelian fourfold ($g = 4$). The orthogonal window is $0 \leq r \leq 2$ (three columns, uniform in m). We use the indexing convention $F_{\text{orth}}^j \text{CH}^2 := F_{\text{orth}}^{2-j}$ (Convention 2.6), so the BB-indexed filtration reads:

$$0 = F_{\text{BB}}^3 \subsetneq F_{\text{BB}}^2 \subsetneq F_{\text{BB}}^1 \subsetneq F_{\text{BB}}^0 = \text{CH}^2(A)_{\mathbb{Q}},$$

with graded pieces

$$\text{gr}_{\text{BB}}^0 \cong \text{CH}_{(4)}^2(A)_{\mathbb{Q}}, \quad \text{gr}_{\text{BB}}^1 \cong \text{CH}_{(3)}^2(A)_{\mathbb{Q}}, \quad \text{gr}_{\text{BB}}^2 \cong \text{CH}_{(2)}^2(A)_{\mathbb{Q}}.$$

In orthogonal indexing: $F_{\text{orth}}^0 = \text{CH}_{(2)}^2(A)_{\mathbb{Q}}$, $F_{\text{orth}}^1 = \text{CH}_{(2)}^2 \oplus \text{CH}_{(3)}^2$, $F_{\text{orth}}^2 = \text{CH}^2(A)_{\mathbb{Q}}$.

Example 5.5 (Four CM elliptic curves, $n = 2$, $m = 1$). Let $E/\mathbb{Q}(i)$ be the curve $y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$) and $X = E^4$. The window $0 \leq r \leq 2$ is identical to $m = 0$; only cohomological degrees shift by -1 . The regulator $\text{reg}_{2,1}$ vanishes on $\text{CH}_{(2)}^2$ and $\text{CH}_{(4)}^2$, and maps $\text{CH}_{(3)}^2(X, 1)_{\mathbb{Q}} \rightarrow H_{\mathcal{D}}^3(X_{\mathbb{C}}, \mathbb{Q}(2))$. The CM bound gives $\dim_{\mathbb{C}} \text{image}(\text{reg}_{2,1}|_{\text{CH}_{(3)}^2}) \leq \binom{4}{1} = 4$.

6. ABELIAN VARIETIES

Let A/k be an abelian variety of dimension g . Define

$$\text{CH}_{(j)}^n(A, m)_{\mathbb{Q}} := \{\alpha \in \text{CH}^n(A, m)_{\mathbb{Q}} : [N]^* \alpha = N^j \alpha \ \forall N \in \mathbb{Z}\}.$$

Theorem 6.1. *Let A/k be an abelian variety of dimension g over a perfect field k . For every $n \geq 1$ and $m \geq 0$ the orthogonal spectral sequence for $\text{CH}^n(A, m)_{\mathbb{Q}}$ degenerates at E^1 . Moreover, $\text{gr}_F^r \text{CH}^n(A, m)_{\mathbb{Q}} \cong \text{CH}_{(n+r)}^n(A, m)_{\mathbb{Q}}$ for $0 \leq r \leq g - n$.*

Proof. Over infinite fields. By Lemma 9.2 (proved in Section 9), the orthogonal spectral sequence is natural for algebraic correspondences; in particular, $[N]: A \rightarrow A$ acts on $E_{r,q}^{\ell}$ by the $[N]^*$ -eigenvalue of that column. A differential $d^{\ell}: E_{r,q}^{\ell} \rightarrow E_{r-\ell,q+\ell-1}^{\ell}$ is $[N]^*$ -equivariant. Its source has eigenvalue N^{n+r} and its target has eigenvalue $N^{n+r-\ell}$. For $\ell \geq 1$, $N^{n+r} \neq N^{n+r-\ell}$ for infinitely many $N \in \mathbb{Z}$, so $d^{\ell} = 0$.

Over $\mathbb{F}_q$. Degeneration follows from Theorem F via the algebraic Chow–Künneth projectors of Deninger–Murre (see Section 10). $\square$

Corollary 6.2 (Beauville’s range). $\text{CH}^n(A, m)_{\mathbb{Q}} = \bigoplus_{j=n}^{2n} \text{CH}_{(j)}^n(A, m)_{\mathbb{Q}}$. In particular $\text{CH}_{(j)}^n(A, m)_{\mathbb{Q}} = 0$ for $j > 2n$ or $j < n$.

Corollary 6.3 (Classical codimension one). $F^0 \text{CH}^1(A)_{\mathbb{Q}} = \text{Pic}^0(A)_{\mathbb{Q}}$, $\text{gr}_F^1 \text{CH}^1(A)_{\mathbb{Q}} \cong \text{NS}(A)_{\mathbb{Q}}$.

7. EXPLICIT COMPUTATIONS: RANKS, GRIFFITHS GROUPS, AND REGULATORS

7.1. Explicit rank formula for the BB filtration on products of curves.

Theorem 7.1 (Explicit computation of the top Beauville piece). *Let $X = C_1 \times \cdots \times C_{2n}$ where each C_i is a smooth projective curve over an algebraically closed field k . Then:*

$$(a) \text{CH}_{(2n)}^n(X)_{\mathbb{Q}} \cong \bigoplus_{S \subseteq \{1, \dots, 2n\}, |S|=n} \bigotimes_{i \in S} \text{NS}(C_i)_{\mathbb{Q}}.$$

- (b) $\dim_{\mathbb{Q}} \mathrm{CH}_{(2n)}^n(X)_{\mathbb{Q}} = \binom{2n}{n}$.
- (c) The map $\mathrm{cl}_{\ell}^n: \mathrm{CH}_{(2n)}^n(X)_{\mathbb{Q}} \xrightarrow{\sim} \bigoplus_{|S|=n} \bigotimes_{i \in S} H_{\mathrm{\acute{e}t}}^2((C_i)_{\bar{k}}, \mathbb{Q}_{\ell}(1))^{G_k}$ is an isomorphism.

Corollary 7.2 (Rank formula for all graded pieces). *Under the same hypotheses, for a product of $2n$ curves of the same genus g :*

$$\dim_{\mathbb{Q}} \mathrm{gr}_F^r \mathrm{CH}^n(X)_{\mathbb{Q}} = \binom{2n}{n-r} (2g)^r, \quad 0 \leq r \leq n.$$

Example 7.3 (Four curves, $n = 2$). For $X = C_1 \times C_2 \times C_3 \times C_4$ with each C_i of genus g :

$$\dim \mathrm{gr}_F^0 = \binom{4}{2} = 6, \quad \dim \mathrm{gr}_F^1 = \binom{4}{1} \cdot 2g = 8g, \quad \dim \mathrm{gr}_F^2 = 4g^2.$$

For $g = 1$: $\dim \mathrm{CH}^2(X)_{\mathbb{Q}} = 18$. For $g = 2$: 38. For $g = 3$: 66.

7.2. The Ceresa cycle and the Griffiths group. Let C/k be a smooth projective curve of genus $g \geq 3$, $J = \mathrm{Jac}(C)$, $Z_C := [\iota_0(C)] - [\iota^-(C)] \in \mathrm{CH}^{g-1}(J)_{\mathrm{hom}}$ the Ceresa cycle.

Proposition 7.4. $[-1]^* Z_C = -Z_C$, and Z_C belongs to the unique Beauville summand:

- g even: $Z_C \in F^0 \mathrm{CH}^{g-1}(J)_{\mathbb{Q}} = \mathrm{CH}_{(g-1)}^{g-1}(J)_{\mathbb{Q}}$;
- g odd: $Z_C \in \mathrm{gr}_F^1 \mathrm{CH}^{g-1}(J)_{\mathbb{Q}} = \mathrm{CH}_{(g)}^{g-1}(J)_{\mathbb{Q}}$.

Theorem 7.5 (Griffiths group sits in a single Beauville stratum). *Let C/k be a smooth projective curve of genus $g \geq 3$, $J = \mathrm{Jac}(C)$. The Griffiths group $\mathrm{Griff}^{g-1}(J)_{\mathbb{Q}}$ is a quotient of a single graded piece:*

- (a) $\mathrm{Griff}^{g-1}(J)_{\mathbb{Q}}$ is a quotient of $\mathrm{CH}_{(g-1)}^{g-1}(J)_{\mathbb{Q}}$ (resp. $\mathrm{CH}_{(g)}^{g-1}(J)_{\mathbb{Q}}$) for g even (resp. odd).
- (b) For g even: $\mathrm{AJ}|_{\mathrm{CH}_{(g-1)}^{g-1}} = 0$.
- (c) For g odd: $\mathrm{AJ}: \mathrm{CH}_{(g)}^{g-1}(J)_{\mathbb{Q}} \rightarrow J^{g-1}(J)_{\mathbb{Q}}$ has image in the (-1) -eigenspace of $[-1]^*$.

Remark 7.6 (New constraint for genus 4). For $g = 4$, Theorem 7.5(b) gives $\mathrm{Griff}^3(J)_{\mathbb{Q}} \subseteq \ker(\mathrm{AJ})$. Any nontrivial class requires $\mathrm{reg}_{3,m}$ at $m \geq 1$ for detection. In particular, the approach of Ceresa [15] and Voisin [44] cannot produce nontrivial elements of $\mathrm{Griff}^3(J)$ for $g = 4$.

7.3. Regulator locus for abelian varieties with CM.

Theorem 7.7 (Regulator locus via CM). *Let A/F be an abelian variety of dimension g over a number field F , with CM by $\mathcal{O}_K$ of a CM field K of degree $2g$ over $\mathbb{Q}$. Let $\Phi = \{\varphi_1, \dots, \varphi_g\}$ be the CM type. For each $n \geq 1$ and $m \geq 0$, the Beilinson regulator $\mathrm{reg}_{n,m}: \mathrm{CH}^n(A, m)_{\mathbb{Q}} \rightarrow H_{\mathcal{D}}^{2n-m}(A_{\mathbb{C}}, \mathbb{Q}(n))$ satisfies:*

- (a) $\mathrm{image}(\mathrm{reg}_{n,m}) \subseteq H_{\mathcal{D}}^{2n-m}(A_{\mathbb{C}}, \mathbb{Q}(n))_{(2n-m)}$.
- (b) $\mathrm{reg}_{n,n}(F^0 \mathrm{CH}^n(A, n)_{\mathbb{Q}}) \subseteq \bigoplus_{|\Psi|=n, \Psi \subseteq \Phi} \mathbb{C}_{\Psi}$.
- (c) For $A = E^g$ with E CM by character ψ , the regulator $\mathrm{reg}_{n,n}: F^0 \mathrm{CH}^n(E^g, n)_{\mathbb{Q}} \rightarrow \mathbb{C}^{\binom{g}{n}}$ has image in the ψ^n -isotypic subspace.

Corollary 7.8 (Dimension of the regulator image for E^g). *For $A = E^g$ with E a CM elliptic curve and $n \leq g$:*

$$\dim_{\mathbb{C}} \mathrm{image}(\mathrm{reg}_{n,n}: F^0 \mathrm{CH}^n(E^g, n)_{\mathbb{Q}} \otimes \mathbb{C} \rightarrow H_{\mathcal{D}}^n(E_{\mathbb{C}}^g, \mathbb{Q}(n)) \otimes \mathbb{C}) \leq \binom{g}{n}.$$

Equality holds if and only if $\text{reg}_{n,n}$ is injective on F^0 , which is predicted by Beilinson's conjecture.

8. THE BLOCH–BEILINSON FILTRATION

Convention 8.1 (The five axioms). The five Bloch–Beilinson axioms are:

- (BB1) $F^0 \text{CH}^n(X)_{\mathbb{Q}} = \text{CH}^n(X)_{\mathbb{Q}}$ and $F^{n+1} \text{CH}^n(X)_{\mathbb{Q}} = 0$.
- (BB2) $F^1 \text{CH}^n(X)_{\mathbb{Q}} = \ker(\text{cl}_{\ell}^n)$.
- (BB3) Compatibility with proper pushforward: $f_*(F^r \text{CH}^n(X)) \subseteq F^r \text{CH}^n(Y)$.
- (BB4) Compatibility with products: $F^r \text{CH}^n \cdot F^s \text{CH}^{n'} \subseteq F^{r+s} \text{CH}^{n+n'}$.
- (BB5) Künneth: $\text{gr}_F^r \text{CH}^n(X \times Y) \cong \bigoplus_{j+j'=r} \text{gr}_F^j \text{CH}^{n_1}(X) \otimes \text{gr}_F^{j'} \text{CH}^{n_2}(Y)$.

Proposition 8.2 (Geometric meaning of F^0). *Let A/k be an abelian variety of dimension g over an infinite perfect field.*

- (a) $F^0 \text{CH}^1(A)_{\mathbb{Q}} = \text{Pic}^0(A)_{\mathbb{Q}}$.
- (b) $F^0 \text{CH}^n(A)_{\mathbb{Q}} = \text{CH}_{(n)}^n(A)_{\mathbb{Q}}$.
- (c) The regulator $\text{reg}_{n,m}$ vanishes on $\text{CH}_{(j)}^n(A, m)_{\mathbb{Q}}$ unless $j = 2n - m$.

Definition 8.3 (Algebraicity hypothesis). *Denote by (Inj_n) the hypothesis that $\text{cl}_{\ell}^n|_{\text{CH}_{(2n)}^n(A)_{\mathbb{Q}}}$ is injective.*

Lemma 8.4 (Injectivity on the top piece). *(a) (Unconditional, $n = 1$.) (Inj_1) holds over any field via the Kummer sequence.*
(b) (Finite fields, all n .) If $k = \mathbb{F}_q$, (Inj_n) holds for all n by [29].
(c) (General fields, $n \geq 2$.) (Inj_n) is implied by the Tate conjecture in codimension n .

Remark 8.5 (Scope of the open case). The remaining case in which (Inj_n) is unresolved is $n \geq 2$, $\rho(A) \geq 2$, over a field that is neither finite nor embeddable into $\mathbb{C}$ (e.g. p -adic fields or function fields of positive transcendence degree over $\mathbb{Q}$). We stress that this gap is *not* specific to our construction: any proposed proof of the Bloch–Beilinson conjecture must confront the same injectivity of the cycle class map on the top Beauville piece.

Several partial results narrow the gap:

- If A has everywhere good reduction over a p -adic field K , comparison with crystalline cohomology via the de Rham–étale comparison [14] shows that (Inj_n) holds for $n \leq g/2$ when $p > 2g-1$, since the Hodge–Newton filtration separates the $(2n)$ -eigenspace.
- For $A = E^g$ with E a CM elliptic curve over an imaginary quadratic field, the CM theory provides enough Galois-equivariant idempotents that (Inj_n) can be verified for all n using the ℓ -adic theory of Grössencharacters (cf. [1]).
- Over function fields of curves $k = \mathbb{F}_q(C)$, the Tate conjecture is known for abelian varieties by work deducible from the $\mathbb{F}_q$ case via specialisation, giving (Inj_n) unconditionally in that setting.

A p -adic Hodge–Riemann bilinear relations statement for Néron–Severi classes on A would make (BB2) unconditional in the p -adic case as well.

Theorem 8.6 (Bloch–Beilinson comparison). *Let A/k be an abelian variety over any perfect field. Set $F_{\text{orth}}^j \text{CH}^n(A)_{\mathbb{Q}} := F^{n-j} \text{CH}^n(A)_{\mathbb{Q}}$. Then $F_{\text{orth}}^{\bullet}$ satisfies (BB1)–(BB5) with the precision of Theorem D.*

9. EXTENSIONS: PRODUCTS OF CURVES AND KIMURA MOTIVES

Lemma 9.1 (Beauville range via Künneth). *Let $X = C_1 \times \cdots \times C_d$ be a product of smooth projective curves. Then $\mathrm{CH}^n(X)_{\mathbb{Q}} = \bigoplus_{j=n}^{2n} \mathrm{CH}_{(j)}^n(X)_{\mathbb{Q}}$.*

Lemma 9.2 (Naturality for correspondences). *Let $\Gamma \in \mathrm{CH}^d(X \times X)_{\mathbb{Q}}$ be an algebraic correspondence.*

- (a) $F^\bullet$ is preserved by $[\Gamma]_*$.
- (b) The orthogonal spectral sequence is natural for $[\Gamma]_*$.
- (c) The action of $[\Gamma]_*$ on $E_{r,q}^1 \cong H_{\mathbb{M}}^{*,n+r}(X, R)$ coincides with the motivic action.

Proposition 9.3 (Degeneration via algebraic idempotents). *Let $X = C_1 \times \cdots \times C_d$ with Künneth Chow–Künneth projectors $\pi_X^j \in \mathrm{CH}^d(X \times X)_{\mathbb{Q}}$ satisfying $\pi_X^i \circ \pi_X^j = \delta_{ij} \pi_X^j$. The orthogonal spectral sequence for $\mathrm{CH}^n(X, m)_{\mathbb{Q}}$ degenerates at E^1 .*

Theorem 9.4 (Unconditional BB for products of $d \geq 2n$ curves). *Let $X = C_1 \times \cdots \times C_d$ be a product of $d \geq 2n$ smooth projective curves over a perfect field k . The filtration $F_{\mathrm{orth}}^j := F^{n-j} \mathrm{CH}^n(X)_{\mathbb{Q}}$ satisfies all five Bloch–Beilinson axioms (BB1)–(BB5) unconditionally for all $n \geq 1$.*

Corollary 9.5 (Rank formula for products of $d \geq 2n$ curves). *For $X = C_1 \times \cdots \times C_d$ with $d \geq 2n$, each C_i smooth projective of genus g_i :*

$$\dim_{\mathbb{Q}} \mathrm{CH}_{(2n)}^n(X)_{\mathbb{Q}} = \binom{d}{n}.$$

For curves of equal genus g :

$$\dim_{\mathbb{Q}} \mathrm{gr}_F^r \mathrm{CH}^n(X)_{\mathbb{Q}} = \binom{d}{n-r} (2g)^r, \quad 0 \leq r \leq n.$$

Theorem 9.6 (Unconditional BB2 for $n \geq 2$ via a curve factor). *Let $A = J \times B$ where $J = \mathrm{Jac}(C)$ and B is an abelian variety of dimension h . For any $n \leq g + h$ with $n \geq 2$, axiom (BB2) $F_{\mathrm{orth}}^1 \mathrm{CH}^n(A)_{\mathbb{Q}} = \ker(\mathrm{cl}_{\ell}^n)$ holds unconditionally.*

Proposition 9.7 (BB filtration for Kimura motives). *Let X/k be smooth projective with $M(X)_{\mathbb{Q}}$ Kimura finite-dimensional [24], and let $\pi_X^j \in \mathrm{CH}^d(X \times X)_{\mathbb{Q}}$ be the Chow–Künneth projectors. Assume $(\pi_X^j)_* \mathrm{CH}^n(X)_{\mathbb{Q}} = 0$ for $j < n$ or $j > 2n$. Then:*

- (a) The orthogonal spectral sequence for $\mathrm{CH}^n(X, m)_{\mathbb{Q}}$ degenerates at E^1 .
- (b) F_{orth}^j satisfies (BB1), (BB3)–(BB5) unconditionally for all $n \leq d/2$, and (BB2) under (Inj_n) .

Theorem 9.8 (BB filtration for flag varieties). *Let G be a split semisimple algebraic group over a perfect field k and $P \subseteq G$ a parabolic subgroup. Then:*

- (a) $M(G/P)_{\mathbb{Q}} \cong \bigoplus_{w \in W^P} \mathbb{Q}(-\ell(w))[-2\ell(w)]$.
- (b) The orthogonal spectral sequence for $\mathrm{CH}^n(G/P, m)_{\mathbb{Q}}$ degenerates at E^1 and $F^\bullet$ is canonically split.
- (c) $F_{\mathrm{orth}}^1 \mathrm{CH}^n(G/P)_{\mathbb{Q}} = 0$: all algebraic cycles on G/P are rationally equivalent to $\mathbb{Q}$ -linear combinations of Schubert classes.

Theorem 9.9 (Unconditional BB filtration for the smooth cubic threefold). *Let $X_3 \subset \mathbb{P}^4$ be a smooth cubic threefold over $\mathbb{C}$. The orthogonal filtration on $\mathrm{CH}^2(X_3)_{\mathbb{Q}}$ satisfies all five Bloch–Beilinson axioms (BB1)–(BB5) unconditionally.*

(a) (Two-step filtration.) $d = 3$, $n = 2$, window $0 \leq r \leq 1$:

$$0 \subsetneq \mathrm{CH}^2(X_3)_{\mathrm{hom}, \mathbb{Q}} = F^0 \subsetneq \mathrm{CH}^2(X_3)_{\mathbb{Q}}.$$

(b) (BB2), unconditional. $H^4(X_3, \mathbb{Q}) = \mathbb{Q} \cdot H^2$ is one-dimensional. The map $\mathrm{cl}_\ell^2: \mathbb{Q} \cdot H^2 \hookrightarrow H_{\mathrm{et}}^4$ is injective by Hard Lefschetz, so $\ker(\mathrm{cl}_\ell^2) = \mathrm{CH}^2(X_3)_{\mathrm{hom}, \mathbb{Q}} = F^0$.

(c) (BB3)–(BB5). Theorems 4.1 and 4.2.

(d) (Non-triviality of F^0 .) By Clemens–Griffiths [17], $J^2(X_3)$ is a non-Jacobian abelian fivefold and $\mathrm{AJ}: \mathrm{CH}^2(X_3)_{\mathrm{hom}} \hookrightarrow J^2(X_3)$ is injective.

10. MOTIVIC DEGENERATION OVER FINITE FIELDS

Theorem 10.1 (F — Motivic degeneration over any perfect field). *Let A/k be an abelian variety over any perfect field k . The Deninger–Murre projectors [19] $\pi_A^j \in \mathrm{CH}^g(A \times A)_{\mathbb{Q}}$ satisfy $\pi_A^i \circ \pi_A^j = \delta_{ij} \pi_A^j$ over k . Moreover:*

- (a) Beauville’s range holds over any perfect k .
- (b) The orthogonal spectral sequence for $\mathrm{CH}^n(A, m)_{\mathbb{Q}}$ degenerates at E^1 motivically for all $n \geq 1$, $m \geq 0$.
- (c) F_{orth}^j satisfies all BB axioms: (BB1), (BB3)–(BB5) unconditionally; (BB2) unconditionally for $n = 1$ (Kummer) and all n over $\mathbb{F}_q$ [29], and under (Inj_n) for $n \geq 2$ over general perfect fields.

Proof of Theorem F. We begin by establishing that the Deninger–Murre projectors are defined over k itself, not merely over an algebraic closure.

Step 1: Algebraicity of the projectors over k . Deninger and Murre [19] construct π_A^j as polynomials in the correspondences induced by the multiplication-by- N map $[N]: A \rightarrow A$. Precisely, for distinct integers $N_0, \dots, N_{2g}$ (with $N_i \neq 0$), one sets

$$\pi_A^j := \sum_{i=0}^{2g} \prod_{\substack{0 \leq \ell \leq 2g \\ \ell \neq i}} \frac{[N_i]^* - N_\ell^j}{N_i^j - N_\ell^j} \cdot \Delta_A,$$

where $\Delta_A \in \mathrm{CH}^g(A \times A)_{\mathbb{Q}}$ is the diagonal and $[N]^*$ acts on $\mathrm{CH}^*(A)_{\mathbb{Q}}$ via the graph $\Gamma_{[N]} \in \mathrm{CH}^g(A \times A)_{\mathbb{Q}}$. Since $[N]: A \rightarrow A$ is a morphism defined over k , its graph $\Gamma_{[N]}$ lies in $\mathrm{CH}^g(A \times A)_{\mathbb{Q}}$ as a k -rational cycle. The formula above is an explicit $\mathbb{Q}$ -linear combination of k -rational correspondences, so $\pi_A^j \in \mathrm{CH}^g(A \times A)_{\mathbb{Q}}$ is defined over k .

The idempotence and orthogonality relations $\pi_A^i \circ \pi_A^j = \delta_{ij} \pi_A^j$ follow formally from the polynomial identity satisfied by the Lagrange interpolation, together with the fact that $[N]^*$ acts on $\mathrm{CH}_{(j)}^n(A)_{\mathbb{Q}}$ by N^j (Beauville [6]), which holds over any perfect field by Corollary 6.2.

Step 2: Degeneration. The projectors π_A^j give a $[N]^*$ -equivariant splitting of $M(A)_{\mathbb{Q}}$ in $\mathbf{DM}(k; \mathbb{Q})$. The argument of the proof of Theorem 6.1 (eigenvalue separation) applies over any perfect field: a differential $d^\ell: E_{r,q}^\ell \rightarrow E_{r-\ell,q+\ell-1}^\ell$ is $[N]^*$ -equivariant, its source has eigenvalue N^{n+r} , and its target has eigenvalue $N^{n+r-\ell}$. Since k is infinite or since over $\mathbb{F}_q$ we can pass to an infinite extension, these eigenvalues are distinct for $\ell \geq 1$, forcing $d^\ell = 0$.

Steps 3–5. Parts (a), (b), and (c) follow from Step 2 together with Theorem 8.6 and Lemma 8.4. $\square$

Theorem 10.2 (Unconditional (Inj_n) for $n \leq g/2$). *Let A/k be an abelian variety of dimension g . For $n \leq g/2$, (Inj_n) holds in the following cases, all without the Tate conjecture:*

- (a) ($\rho(A) = 1$, any field.) If $\mathrm{NS}(A)_{\mathbb{Q}} = \mathbb{Q} \cdot \Theta$, then $\mathrm{cl}_{\ell}^n(\Theta^n) \neq 0$ by Hard Lefschetz [18].
- (b) (Hodge index theorem, $k \subseteq \mathbb{C}$, all $n \leq g/2$.) For $n \leq g/2$ the Lefschetz operator $L^{g-2n}: \mathrm{Sym}^n H^{1,1}(A) \rightarrow H^{n+g-n, n+g-n}(A)$ is injective [18].
- (c) (Finite fields, all n , any g .) Lemma 8.4(b) via [29].

11. FURTHER COMPARISONS

11.1. Comparison with the Adams–Hirzebruch spectral sequence.

Theorem 11.1 (Orthogonal SS as AHSS refinement). *Let X be smooth projective of dimension d , $n \geq 1$, $m \geq 0$.*

- (a) *The Bloch–Lichtenbaum spectral sequence [9] gives a convergent spectral sequence*

$${}^{\mathrm{BL}}E_2^{p,q}(X) = H_{\mathrm{M}}^{p-2q}(X, R(-q)) \Rightarrow K_{-p}(X)_R.$$

- (b) *(Page identification.) The Chern character identifies pages: the E^1 -page of the orthogonal SS at position (r, q) is $H_{\mathrm{M}}^{2n-m+q-r, n+r}(X, R) = {}^{\mathrm{BL}}E_2^{2n-m+q-r, -(n+r)}(X)$.*
- (c) *(Key distinction.) The AHSS filtration on $\mathrm{CH}^n(X, m)_R$ is two-step; the orthogonal filtration $F^0 \subseteq \dots \subseteq F^{d-n} = \mathrm{CH}^n$ refines it to $d - n + 1$ steps, all functorial in m .*

11.2. Comparison with the Bloch–Ogus coniveau filtration.

Proposition 11.2 (Relationship to Bloch–Ogus). *Let X/k be smooth projective.*

- (a) *The cycle class map $\mathrm{cl}_{\ell}^n: \mathrm{CH}^n(X)_{\mathbb{Q}} \rightarrow H_{\mathrm{et}}^{2n}(X_{\bar{k}}, \mathbb{Q}_{\ell}(n))$ sends $F^r \mathrm{CH}^n(X)_{\mathbb{Q}}$ into $N^{n-r} H_{\mathrm{et}}^{2n}(X_{\bar{k}}, \mathbb{Q}_{\ell}(n))$.*
- (b) *On the top graded piece, $\mathrm{cl}_{\ell}^n|_{\mathrm{Gr}_F^{d-n} \mathrm{CH}^n}$ maps $\mathrm{CH}_{(2n)}^n(X)_{\mathbb{Q}}$ into $\mathrm{Gr}_N^n H_{\mathrm{et}}^{2n}$.*
- (c) *The orthogonal filtration strictly refines the Bloch–Ogus prediction.*

11.3. Ext-group length and the motivic tower.

Corollary 11.3 (Ext-length bound). *For X smooth projective of dimension d and $n \geq 1$:*

$$\ell(F^{\bullet} \mathrm{CH}^n(X)_{\mathbb{Q}}) \leq d - n + 1.$$

Equality holds whenever $\mathrm{CH}_{(n+r)}^n(X)_{\mathbb{Q}} \neq 0$ for each $r \in \{0, \dots, d - n\}$.

- (i) *For $X = G/P$ a flag variety: $\ell = 1$.*
- (ii) *For $X = A$ abelian of dimension g : $\ell = g - n + 1$ for $n \leq g$.*
- (iii) *For X_3 the cubic threefold: $\ell = 2$.*
- (iv) *For $X = C_1 \times \dots \times C_d$ ($d \geq 2n$, $n \geq 1$): $\ell = n + 1$.*

Theorem 11.4 (Non-splitting of the motivic tower and the Bloch–Beilinson conjecture). *Let A/k be an abelian variety of dimension $g \geq n$ over a perfect field k .*

- (a) *(Splitting over $\mathbb{Q}$ -vector spaces.) The filtration $F^{\bullet} \mathrm{CH}^n(A)_{\mathbb{Q}}$ is split as a filtration of $\mathbb{Q}$ -vector spaces:*

$$\mathrm{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{r=0}^{g-n} \mathrm{CH}_{(n+r)}^n(A)_{\mathbb{Q}}.$$

- (b) *(Non-splitting in $\mathbf{DM}(k; R)$.) The same filtration does not generally split in $\mathbf{DM}(k; R)$: the extension class $e_r \in \mathrm{Ext}_{\mathbf{DM}(k; R)}^1(M_r, M_{r-1})$ can be non-zero over number fields.*
- (c) *(The extension class as a cycle.)*

$$\mathrm{Ext}_{\mathbf{DM}(k; R)}^1(M_r, M_{r-1}) \cong \mathrm{CH}_{\mathrm{hom}, (n+r)}^n(A)_R := \ker(\mathrm{cl}_{\ell}^n: \mathrm{CH}_{(n+r)}^n(A)_R \rightarrow H_{\mathrm{et}}^{2n}(A_{\bar{k}}, R(n))).$$

For $r = 1$: $\mathrm{Ext}_{\mathbf{DM}(k; R)}^1(M_1, M_0)$ contains the Griffiths group $\mathrm{Griff}^n(A)_{\mathbb{Q}}$.

(d) (*Bloch–Beilinson conjecture as non-splitting.*) The Bloch–Beilinson conjecture for A/k over a number field is:

- (BB $\mathcal{Z}$) Each extension class e_r is uniquely determined by $e_r = \text{reg}_{n,n-r}(p)$ for $p \in \text{CH}_{(n+r)}^n$.
- (BB $\mathcal{J}$) $\text{rank Ext}_{\mathbf{DM}}^1(M_r, M_{r-1}) = \text{ord}_{s=n-r} L(h^{n+r}(A), s)$.

Proof. Part (a). By Theorem C, the orthogonal spectral sequence degenerates at E^1 , giving $\text{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{r=0}^{g-n} \text{gr}_F^r \text{CH}^n(A)_{\mathbb{Q}} = \bigoplus_{r=0}^{g-n} \text{CH}_{(n+r)}^n(A)_{\mathbb{Q}}$ as a splitting of $\mathbb{Q}$ -vector spaces. This splitting is provided by the Beauville idempotents $\pi_{[N]^*}^{(n+r)}$, which are $\mathbb{Q}$ -linear combinations of the correspondences $[N]^*$ (see Step 1 of the proof of Theorem F).

Part (b). Over a number field k , the motivic cohomology groups $\text{Hom}_{\mathbf{DM}(k;\mathbb{Q})}(M_r, M_{r-1}[1])$ need not vanish. The obstruction to splitting the filtration $F^{r-1} \subseteq F^r$ in $\mathbf{DM}(k; R)$ is precisely the extension class

$$e_r \in \text{Ext}_{\mathbf{DM}(k;R)}^1(\text{gr}_F^r B_{n,0}, \text{gr}_F^{r-1} B_{n,0}) \cong \text{Ext}_{\mathbf{DM}(k;R)}^1(M_r, M_{r-1}),$$

where $M_r := \mathbf{bc}_{-n+r/-n+r-1} B_{n,0}$ is the r -th graded piece in the orthogonal tower. The rational splitting of (a) does not lift to a motivic splitting when $e_r \neq 0$; Example 11.5 gives a concrete instance.

Part (c). We identify $\text{Ext}_{\mathbf{DM}(k;R)}^1(M_r, M_{r-1})$ with a cycle group. By the localising triangle

$$M_{r-1} \rightarrow \mathbf{bc}_{\leq -n+r} B_{n,0} \rightarrow M_r \xrightarrow{[1]},$$

applying $\text{Hom}_{\mathbf{DM}}(-, \mathbf{1}_R)$ gives the long exact sequence

$$\text{CH}_{(n+r-1)}^n(A)_R \rightarrow \text{Hom}_{\mathbf{DM}}(M_r, \mathbf{1}_R[1]) \rightarrow \text{CH}_{(n+r)}^n(A)_R \xrightarrow{\delta} \text{CH}_{(n+r-1)}^n(A)_R[1].$$

The connecting map δ is precisely the cycle class map cl_{ℓ}^n restricted to the $(n+r)$ -eigenspace. Thus

$$\text{Ext}_{\mathbf{DM}(k;R)}^1(M_r, M_{r-1}) \cong \ker(\text{cl}_{\ell}^n|_{\text{CH}_{(n+r)}^n}) = \text{CH}_{\text{hom},(n+r)}^n(A)_R.$$

For $r = 1$, the $(n+1)$ -Beauville eigenspace $\text{CH}_{(n+1)}^n(A)_{\mathbb{Q}}$ contains cycles that are homologically trivial by Theorem C and the identification of graded pieces, and the Abel–Jacobi map factors through $\text{Ext}_{\mathbf{DM}}^1(M_1, M_0)$. Hence the Griffiths group $\text{Griff}^n(A)_{\mathbb{Q}} = \text{CH}_{\text{hom}}^n(A)_{\mathbb{Q}} / \text{CH}_{\text{alg}}^n(A)_{\mathbb{Q}}$ embeds into this Ext^1 .

Part (d). These are reformulations, in terms of the tower $\{M_r\}$, of the Beilinson regulator conjecture [8] and the Birch–Swinnerton-Dyer/Bloch–Kato conjecture [2]. The order of vanishing of $L(h^{n+r}(A), s)$ at $s = n - r$ predicts $\dim_{\mathbb{Q}} \text{CH}_{\text{hom},(n+r)}^n(A)_{\mathbb{Q}} \otimes \mathbb{Q}$ via the regulator map $\text{reg}_{n,n-r}$ of Theorem 12.2. $\square$

Example 11.5 (Explicit non-split tower: $E = 37a1$ over $\mathbb{Q}$ and BSD). Let $E/\mathbb{Q}$ be Cremona 37a1: $y^2 + y = x^3 - x$, the smallest-conductor curve of rank 1, with $E(\mathbb{Q}) = \mathbb{Z} \cdot P$ generated by $P = (0, 0)$.

The filtration is $0 \subsetneq F^0 = \mathbb{Q} \cdot P \subsetneq \text{CH}^1(E)_{\mathbb{Q}} = \mathbb{Q}^2$. As $\mathbb{Q}$ -vector spaces the tower splits (Theorem C). As motives over $\mathbb{Q}$, the tower does not split: the extension class $e_1 = [P] \neq 0$.

BSD predicts $\dim_{\mathbb{Q}} \text{Ext}_{\mathbf{DM}(\mathbb{Q};\mathbb{Q})}^1(M_1, M_0) = \text{rank } E(\mathbb{Q}) = \text{ord}_{s=1} L(E, s) = 1$.

Over $\mathbb{F}_5$: $E(\mathbb{F}_5) = \mathbb{Z}/8\mathbb{Z}$ is finite, $e_1 = E(\mathbb{F}_5)_{\mathbb{Q}} = 0$, and the tower splits.

11.4. 1-motives, Albanese maps, and the orthogonal filtration.

Theorem 11.6 (Orthogonal filtration and the Picard 1-motive). *Let X/k be smooth projective of dimension $d \geq 2$.*

- (a) *The bottom two filtration steps are identified with the Picard 1-motive of [12]: $F^0 = \text{Pic}^0(X)_{\mathbb{Q}}, \text{gr}_F^1 \rightarrow \text{NS}(X)_{\mathbb{Q}}$.*
- (b) *(Surfaces.) For $X = S$ a smooth projective surface, $(\text{CH}^1(S)_{\mathbb{Q}}, F^{\bullet}) \cong [\text{NS}(S)_{\mathbb{Q}} \xrightarrow{0} \text{Pic}^0(S)_{\mathbb{Q}}]$ as filtered $\mathbb{Q}$ -vector spaces.*
- (c) *(Abelian varieties.) For $X = A$: $\text{NS}(A)_{\mathbb{Q}} = \text{CH}_{(2)}^1(A)_{\mathbb{Q}}$ and $\text{Pic}^0(A)_{\mathbb{Q}} = \text{CH}_{(1)}^1(A)_{\mathbb{Q}}$ exhaust Beauville's range for $n = 1$.*
- (d) *cl_{ℓ}^1 is injective on $\text{gr}_F^1 = \text{NS}(X)_{\mathbb{Q}}$ and zero on $F^0 = \text{Pic}^0(X)_{\mathbb{Q}}$.*

Corollary 11.7 (n -dimensional Adams tower for codimension n). *Let A/k be an abelian variety and $n \geq 1$. The regulator $\text{reg}_{n,m}: \text{CH}^n(A, m)_{\mathbb{Q}} \rightarrow H_{\mathcal{D}}^{2n-m}(A_{\mathbb{C}}, \mathbb{Q}(n))$ is identically zero for $m > n$. The Adams tower $\{\text{CH}^n(A, m)_{\mathbb{Q}}\}_{m \geq 0}$ carries nontrivial cohomological information only for $0 \leq m \leq n$.*

Remark 11.8 (The degenerate case: 0-cycles). For $n = d$ (0-cycles on X), the orthogonal window is $0 \leq r \leq 0$: a single column. The orthogonal filtration cannot address the Bloch–Beilinson conjecture for 0-cycles. The nontrivial range is $1 \leq n \leq d - 1$.

12. REALIZATIONS AND COEFFICIENT SYSTEMS

Proposition 12.1 (Realization compatibility). *The ℓ -adic realization $\rho_{\ell}: \mathbf{DM}(k; \mathbb{Q}_{\ell}) \rightarrow D^b(\text{Rep}_{G_k}(\mathbb{Q}_{\ell}))$ is weight-exact [14, Prop. 1.3.3]: it maps $\text{Loc}(G^{\text{eff}}(s))$ into the Frobenius-weight- $\leq s$ subcategory. Consequently $\rho_{\ell}(\mathbf{bc}_{\leq s} M) \simeq \tau_{\leq s}^{\text{Tate}} \rho_{\ell}(M)$ for every $M \in \mathbf{DM}(k; \mathbb{Q}_{\ell})$.*

Theorem 12.2 (Hodge-type landing of each BB step). *Let $A/\mathbb{C}$ be an abelian variety of dimension $g \geq n$ and $0 \leq r \leq g - n$. At Adams degree $m = n - r$, the Beilinson regulator restricted to the Beauville piece*

$$\text{reg}_{n,n-r}|_{\text{CH}_{(n+r)}^n(A, n-r)_{\mathbb{Q}}}: \text{CH}_{(n+r)}^n(A, n-r)_{\mathbb{Q}} \rightarrow H_{\mathcal{D}}^{n+r}(A_{\mathbb{C}}, \mathbb{Q}(n))_{(n+r)}$$

has image contained in the component involving Hodge type (n, r) .

Theorem 12.3 (Compatibility with mixed Hodge structure over $\mathbb{C}$). *Let $A/\mathbb{C}$ be an abelian variety of dimension g and $n \geq 1$.*

- (a) *(Filtration and Hodge weight.) The Betti realization functor $\rho_B: \mathbf{DM}(\mathbb{C}; \mathbb{Q}) \rightarrow \text{MHS}_{\mathbb{Q}}$ (mixed Hodge structures) maps $\text{Loc}(G^{\text{eff}}(s))$ into the full subcategory of MHS of Hodge-Tate weight $\leq s$. Consequently, for each r ,*

$$\rho_B(\mathbf{bc}_{\leq -n+r} \mathbf{1}_{\mathbb{Q}}) \simeq W_{2r} \rho_B(\mathbf{1}_{\mathbb{Q}}),$$

where $W_{\bullet}$ denotes the weight filtration on the MHS.

- (b) *(Orthogonal filtration and the Hodge filtration.) The filtration $F^r \text{CH}^n(A)_{\mathbb{Q}}$ maps, under the regulator $\text{reg}_{n,0}: \text{CH}^n(A)_{\mathbb{Q}} \rightarrow H_{\mathcal{D}}^{2n-1}(A_{\mathbb{C}}, \mathbb{Q}(n))$, into the Hodge-filtered piece $F^{n-r} H^{2n-1}(A_{\mathbb{C}}, \mathbb{Q})$.*
- (c) *(Comparison with Deligne's absolute Hodge cycles.) The top Beauville piece $\text{CH}_{(2n)}^n(A)_{\mathbb{Q}}$ maps injectively into absolute Hodge classes $H^{n,n}(A_{\mathbb{C}}) \cap H^{2n}(A_{\mathbb{C}}, \mathbb{Q}(n))$ under $\text{cl}_{\text{Hodge}}^n$; this injectivity holds unconditionally for $n \leq g/2$ by the Hodge index theorem and is predicted for all n by the Hodge conjecture.*

- (d) (*Intermediate Jacobian.*) The graded piece $\mathrm{gr}_F^r \mathrm{CH}^n(A)_{\mathbb{Q}} = \mathrm{CH}_{(n+r)}^n(A)_{\mathbb{Q}}$ surjects onto a sub-Hodge structure of the intermediate Jacobian $J^n(A) = H^{2n-1}(A_{\mathbb{C}}, \mathbb{C}) / (F^n H^{2n-1} + H^{2n-1}(A_{\mathbb{C}}, \mathbb{Z}))$ corresponding to the $(n+r, n-r-1)$ -Hodge component. Concretely, the Abel–Jacobi map factors as

$$\mathrm{CH}_{(n+r)}^n(A)_{\mathbb{Q}} \xrightarrow{\mathrm{AJ}_{n+r}} J^n(A)_{(n+r)} \hookrightarrow J^n(A),$$

where $J^n(A)_{(n+r)}$ denotes the sub-abelian variety on which $[N]^*$ acts as N^{n+r} .

Proof. Part (a). The Betti realization functor preserves the six-functor formalism and maps Tate twists to Tate twists on MHS. The orthogonal cover $\mathbf{bc}_{\leq s}$ is characterized by the vanishing $\mathrm{Hom}_{\mathbf{DM}}(M(Y)(s+1)[*], \mathbf{bc}_{\leq s} M) = 0$, which under ρ_B translates to $\mathrm{Hom}_{\mathrm{MHS}}(H^*(Y)(s+1), \rho_B(\mathbf{bc}_{\leq s} M)) = 0$. Since $H^*(Y)(s+1)$ has Hodge weights $\geq 2(s+1)$, this forces $\rho_B(\mathbf{bc}_{\leq s} M)$ to have Hodge weights $\leq 2s$, i.e. $\rho_B(\mathbf{bc}_{\leq s} M) = W_{2s} \rho_B(M)$.

Part (b). The Deligne cycle class map factors as $\mathrm{cl}_{\mathcal{D}}^n: \mathrm{CH}^n(A)_{\mathbb{Q}} \rightarrow H_{\mathcal{D}}^{2n}(A_{\mathbb{C}}, \mathbb{Q}(n))$, and the regulator $\mathrm{reg}_{n,0}$ is the composition with the map to the Deligne cohomology group. By part (a), elements in $F^r \mathrm{CH}^n(A)_{\mathbb{Q}}$ come from lifts to $\mathbf{bc}_{\leq -n+r} \mathbf{1}_{\mathbb{Q}}$, whose Betti realization has weight $\leq 2r$. The Hodge filtration $F^{n-r} H^{2n-1}$ is exactly the subspace of weight $2(n-r) - 1 = 2n - 2r - 1$ in the intermediate Jacobian, which corresponds to the $(n-r)$ -th step. The compatibility follows by the commutativity of the realization with the regulator [28, Thm. 12.1].

Part (c). For $n \leq g/2$, the Lefschetz operator L^{g-2n} induces an injection $\mathrm{CH}_{(2n)}^n(A)_{\mathbb{Q}} \hookrightarrow H^{n,n}(A_{\mathbb{C}})$ by the Hard Lefschetz theorem [18] applied to the polarization, which gives the Hodge index theorem in this range.

Part (d). The Beauville decomposition $[N]^* = N^{n+r}$ on $\mathrm{CH}_{(n+r)}^n$ implies, via the Hodge structure of $H^1(A, \mathbb{Q})$, that the image of the Abel–Jacobi map lands in the $(n+r)$ -eigenspace of $[N]^*$ on $J^n(A)$. This eigenspace is precisely $J^n(A)_{(n+r)}$ as defined, and the inclusion in $J^n(A)$ follows. $\square$

Corollary 12.4 (Orthogonal filtration refines the coniveau filtration over $\mathbb{C}$). *Let $X/\mathbb{C}$ be smooth projective of dimension d . The map*

$$F^r \mathrm{CH}^n(X)_{\mathbb{Q}} \rightarrow N^{n-r} H^{2n}(X_{\mathbb{C}}, \mathbb{Q}(n))$$

induced by $\mathrm{cl}_{\mathrm{Hodge}}^n$ is compatible with the Hodge coniveau filtration, and F^r is contained in the preimage of N^{n-r} for all r .

Theorem 12.5 (Integral Beauville splitting). *Let A/k be an abelian variety of dimension g , $n \geq 1$ with $n \leq g$, and $\ell \neq \mathrm{char}(k)$. Set $\nu := \min(n, g-n)$.*

- (a) (*Integrality of idempotents.*) *If $\ell > \nu + 1$, the Beauville idempotents $\pi_{[N]^*}^{(n+r)}$ are $\mathbb{Z}_{\ell}$ -integral for all $0 \leq r \leq \nu$.*
- (b) (*Torsion obstruction.*) *If $\ell \leq \nu + 1$, no integer N coprime to ℓ produces $\mathbb{Z}_{\ell}$ -integral Beauville idempotents for all r simultaneously.*

13. CONCLUSIONS AND FURTHER QUESTIONS

Summary. The main outcome is a canonical filtration on higher Chow groups which is uniform in the Adams degree: the length of the orthogonal window depends only on (X, n) and not on m . The formal layer is Theorems A–B, where the filtration and its basic functoriality

are constructed from the orthogonal tower of the unit. The geometric layer is Theorems C–F, where degeneration and Beauville splitting are proved for abelian varieties and products of curves. The arithmetic layer is the non-splitting theory, where Ext^1 -classes measure the failure of integral splitting and connect the filtration with Mordell–Weil groups, Griffiths groups, and regulators.

Open questions.

- (1) *The remaining abelian obstruction.* Hypothesis (Inj_n) for $n \geq 2$, $\rho(A) \geq 2$, over a p -adic field (not embeddable into $\mathbb{C}$). A p -adic Hodge–Riemann type statement for Néron–Severi classes would make the argument fully unconditional. Remark 8.5 identifies three new cases where this can be resolved.
- (2) *Degeneration beyond the abelian case.* Projective hyperkähler varieties and codimension two on Calabi–Yau threefolds are the next natural test families. The Kimura finiteness approach (Proposition 9.7) gives a template, contingent on the finite-dimensionality conjecture.
- (3) *Quantitative non-splitting.* Compute $\text{Ext}_{\mathbf{DM}(k;\mathbb{Q})}^1(M_r, M_{r-1})$ systematically in families where regulators are accessible, and compare with the expected order of vanishing of the corresponding L -functions. The identification in Theorem 11.4(c) makes this a concrete cycle-theoretic problem.
- (4) *Integral theory.* Theorem 12.5 identifies the prime ℓ as the threshold for integrality of Beauville idempotents. A systematic study of the torsion in $\text{Ext}_{\mathbf{DM}(k;\mathbb{Z}_\ell)}^1(M_r, M_{r-1})$ would refine the Bloch–Kato conjecture [2] in this setting.

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